

Strain Selection for the First Isobutanol Single-cell RNA-seq Run

Twelve strains on hand, six to grow

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Twelve strains from one source paper are on hand, and all twelve are *Saccharomyces cerevisiae* CEN.PK2-1C derivatives rather than S288C or BY4741, which decides the reference genome the sequencing is mapped against. Eight of the twelve make isobutanol. Six of those eight ferment 15% glucose and can therefore be grown under one protocol, and they divide into three pairs that differ by a single plasmid, which is what makes a measured difference between two of them attributable. Those six are the first run. Table 1 is everything on hand, Table 2 is the six and Table 3 their genotypes, and Table 4 is what is held back and why. Section 5 carries the one thing that has to be fixed first: the only plasmid-free strain among the six needs a different growth medium than the other five, which would otherwise confound the pair it belongs to.

Tables generated by `experiments/W037-isobutanol-scrnaseq/scripts/strain\_tables.py`

Working notes: `notes/experiments.W037-isobutanol-scrnaseq.strain-selection.md`

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1 Terms ✗

add-back A deleted gene supplied again on a plasmid. In this set the add-back is driven by P_{TDH3} , a strong constitutive promoter, so an add-back strain overexpresses the gene rather than restoring it to its native level. The distinction matters wherever an add-back is read as a stand-in for the wild type, and Section 5 returns to it.

biosensor A two-part construct carried by all twelve strains, not only the isobutanol ones. The first part is the reporter, yEGFP under P_{LEU1} , which Leu3p activates when bound to α -isopropylmalate; what is sensed is therefore that intermediate rather than the alcohol itself. The second part is a leucine-insensitive α -isopropylmalate synthase, either $LEU4^{1-410}$ or $LEU4\Delta S547$, which keeps producing the intermediate under leucine feedback. The synthase is part of the sensor rather than part of a production pathway.

biosensor configuration Which of three constructs a strain carries. The *isobutanol* configuration adds a PEST degradation tag to yEGFP and pairs it with $LEU4^{1-410}$, because in a *LEU2*-deleted background α -isopropylmalate accumulates as a by-product and raises the background signal. The *isopentanol* configuration omits the PEST tag and pairs the reporter with $LEU4\Delta S547$, because *LEU2* is active there and the intermediate is consumed rather than accumulated. A third construct is the reporter alone, carried by the strains that were hosts for the *LEU4* mutant library: their synthase half arrives on the plasmid instead, so the working sensor is split between chromosome and plasmid. Every strain selected for the first run carries the isobutanol configuration, so all six express yEGFP and the reporter is available as a flow-cytometry check before cells are committed to sequencing.

biosensor locus Where the construct sits. For eleven of the twelve it is *HIS3*, which the cassette both interrupts and restores, so a *his3-1* parent becomes histidine prototrophic on integration. JC0053 is the exception and carries its biosensor at *GAL80*, marked with **natMX6**, because its *HIS3* locus already holds the optogenetic circuit. That strain is *gal80* Δ in any case, so the second locus costs it nothing.

CEN plasmid A centromeric plasmid, present at roughly one to two copies per cell, as against a 2μ plasmid at high copy. Every plasmid in this set is CEN and carries *URA3*, so a plasmid-bearing strain is held on medium lacking uracil and a cell that loses the plasmid stops growing on it.

δ -integration Integration into the YARCDelta5 δ sites, which are repeated, so the cassette lands at an unpredictable site and in an unpredictable number of copies. Copy number is therefore a property of the isolate rather than of the design, and the source paper measured it by digital droplet PCR, finding one to two copies in the strains it checked.

high-cell-density fermentation The production assay behind every titer quoted here: outgrowth in synthetic complete medium with 2% glucose, then resuspension at high density in the same medium with 15% glucose, 48 h at 30 °C in a sealed 24-well plate.

stated, derived How a titer is known. A *stated* titer appears in the source paper as a number. A *derived* titer does not: the paper gives only a fold change against a named strain, so the absolute was obtained by dividing, and it is approximate and carries no uncertainty. Derived values are marked † in every table. The underlying absolutes exist only in figure bars and the paper’s Source Data file, which is not held.

titer Product concentration in the medium at the end of a *high-cell-density fermentation*, in mg/L. A titer is comparable to another only when the carbon source matches, and the carbon source is not constant across this set.

2 What is on hand ^x

The twelve come from Zhang et al. 2022, *Nature Communications* 13:270, doi 10.1038/s41467-021-27852-x, citation key zhangBiosensorBranchedchainAmino2022. Values below were read from the primary article and its Supplementary Information, both held as canonical Zotero PDFs whose sha256 are pinned in the generating script.

The host is CEN.PK2-1C, not BY4741. Supplementary Table 1 of the source paper gives the parent as *S. cerevisiae* CEN.PK2-1C, genotype MATa *ura3-52 trp1-289 leu2-3,112 his3-1 MAL2-8c SUC2*. No strain in the paper derives from S288C or BY4741. Two consequences follow. Reads should be mapped against a CEN.PK assembly, because CEN.PK and S288C differ enough that an S288C-only reference loses reads and miscalls variants. And the reference needs explicit entries for the transgenes, since none of them are in any yeast assembly: *yEGFP*, the three *LEU4* alleles, *Bs_alsS*, *Ec_ilvC*^{P2D1-A1}, *Ll_ilvD*, *Ll_adhA*^{RE1}, *EL222*, and the *hphMX*, *kanMX*, *natMX* and *ShBle* markers.

The *LEU4* alleles need care beyond being listed. A strain can be *leu4*Δ and still carry *LEU4* on a plasmid, and the plasmid allele may be full length, truncated at residue 410, or deleted for S547. Reads from a 3'-biased protocol will not separate those alleles unless the reference is built to separate them, and a *leu4*Δ strain carrying a plasmid copy will otherwise appear to express a gene it has lost.

Table 1: The twelve strains on hand, all of them *Saccharomyces cerevisiae* CEN.PK2-1C derivatives. *Titer* is the 48 h high-cell-density value for the product named beside it, and the key below the table says how each value is known. Carbon source differs across the set and is not comparable between rows: the three figures these numbers come from use 15% glucose, 15% galactose, and 2% glucose respectively. The last column marks the six selected for the first run.

id	strain	product	titer (mg/L)	carbon	first run
JC0043	YZy311	isobutanol	~118 [†]	15% glucose	✓
JC0044	YZy313	isobutanol	not reported	15% glucose	✓
JC0045	YZy314	isobutanol	1.6× JC0044	15% glucose	✓
JC0046	YZy148	isopentanol	not reported	15% glucose	
JC0047	YZy148 + LEU4 WT	isopentanol	~201 [†]	15% glucose	
JC0048	YZy148 + LEU4dS547	isopentanol	842 ± 23	15% glucose	
JC0049	YZy452	isobutanol	310 ± 15	15% galactose	
JC0050	YZy454	isobutanol	~73 [†]	15% glucose	✓
JC0051	YZy469	isobutanol	227 ± 13	15% glucose	✓
JC0052	YZy91	isobutanol	not reported	15% glucose	✓
JC0053	YZy502	isobutanol	not reported	2% glucose, dark	
JC0054	SHy134	isopentanol	not reported	15% glucose	

bold stated in the paper as a number; † derived by dividing a stated fold change, so approximate and without uncertainty

Four strains make isopentanol, which splits the set in two. JC0046 through JC0048 and JC0054 produce isopentanol; the remaining eight produce isobutanol. There is no single production ranking across the twelve, and the highest titer on hand, 842 ± 23 mg/L from JC0048, is an isopentanol number.

The paper’s best producers are not on hand. The strains that set the paper’s production ceiling are absent from the twelve: YZy505 at 681 ± 29 mg/L isobutanol, YZy148 carrying *LEU4* mutant 6 at 963 ± 32 mg/L isopentanol,

YZy470 at 443 ± 6 mg/L and YZy312 at 378 ± 10 mg/L. JC0053 is YZy502, the plasmid-free parent of YZy505, and not YZy505 itself. Reaching the paper’s ceiling means obtaining or rebuilding strains beyond this set, which is a separate decision from the one this run makes.

3 The six to grow ✗

Six of the eight isobutanol strains ferment 15% glucose, so one protocol covers them. They form three pairs, and within a pair the two strains share a background and differ only by the plasmid they carry.

Table 2: The six selected, as three axes of one variable each. Within an axis the two strains share a background and differ by the plasmid they carry, so a difference measured between them is attributable to that plasmid. *pathway* names the compartment the heterologous isobutanol route was built in. Every strain here ferments 15% glucose, so all six run under one protocol. *pathway* reads *native* where a strain carries no engineered route and makes isobutanol through endogenous valine and Ehrlich metabolism alone, which is the case for both strains of Axis 1. The key below the table says how each titer is known.

axis	id	strain	deletions	pathway	plasmid	titer (mg/L)
1	JC0052	YZy91	<i>bat1</i> Δ, <i>bat2</i> Δ, <i>ilv6</i> Δ	native	none	not reported
	JC0043	YZy311	<i>bat1</i> Δ, <i>bat2</i> Δ, <i>ilv6</i> Δ	native	ILV6 WT	~118 [†]
2	JC0044	YZy313	<i>bat1</i> Δ	mitochondrial	ILV6 WT	not reported
	JC0045	YZy314	<i>bat1</i> Δ	mitochondrial	ILV6 V110E	1.6× JC0044
3	JC0050	YZy454	<i>ilv3</i> Δ, <i>tma29</i> Δ	cytosolic	Ll_ <i>ilvD</i> WT	~73 [†]
	JC0051	YZy469	<i>ilv3</i> Δ, <i>tma29</i> Δ	cytosolic	Ll_ <i>ilvD</i> I433V	227 ± 13

bold stated in the paper as a number; [†] derived by dividing a stated fold change, so approximate and without uncertainty

Axis 1 is a deletion against its add-back. JC0052 is *bat1*Δ *bat2*Δ *ilv6*Δ with the biosensor and no pathway genes. JC0043 is that same strain carrying *ILV6* on a CEN plasmid. *ILV6* encodes the regulatory subunit of acetolactate synthase, and deleting it relieves valine inhibition, which is why the deletion raises isobutanol in the first place. This pair is the most direct point of contact with knockout-collection data, because the lesion is a single gene whose isobutanol phenotype a deletion screen reports on.

Axis 2 is one allele of one gene, in a producing background. JC0044 and JC0045 are both YZy363, which is *bat1*Δ with a five-cassette mitochondrial isobutanol pathway δ -integrated and *ILV6* intact. They differ in the plasmid allele, wild type against V110E, and the paper reports V110E as 1.6 times the wild-type titer without giving either absolute. Because *ILV6* is intact here, both strains overexpress the gene on top of a native copy, which is a different genetic situation from Axis 1 and not a repeat of it.

Axis 3 is a heterologous enzyme against its improved variant. JC0050 and JC0051 are both YZy452, which is *ilv3*Δ *tma29*Δ with a cytosolic pathway built from *Bs_alsS*, *Ec_ilvC*^{P2D1-A1} and *Ll_ilvD*. They differ in the plasmid allele of *Ll_ilvD*, wild type against I433V, and the variant is 3.1 times the wild type at 227 ± 13 mg/L. This axis carries the largest measured difference of the three and the highest glucose-grown titer in the set. Its two deletions, *ILV3* and *TMA29*, are both genes a deletion screen covers, so it is a second point of contact with knockout data rather than a purely synthetic contrast.

Table 3: Genotypes of the six, as Supplementary Table 1 of the source paper gives them, with the selection markers and the growth medium each one needs. *medium* follows from the plasmid: a strain carrying a URA3 plasmid is held on SC-ura, and JC0052, which carries none, needs uracil instead. Section 5 treats that mismatch, which is the one thing in this list that has to be fixed before the strains are grown.

id	strain and genotype
JC0052	YZy91 $his3::HIS3-P_{LEU1}-yEGFP-PEST-T_{ADH1}-P_{TPI1}-LEU4^{1-410}-T_{PGK1}$, $bat1\Delta::hphMX$, $bat2\Delta::lox71-kanMX-lox66$, $ilv6\Delta::lox71-natMX-lox66$ markers: hphMX, kanMX, natMX, HIS3 medium: SC + uracil
JC0043	YZy311 YZy91, CEN URA3 plasmid ($P_{TDH3}-ILV6-T_{ADH1}$) markers: hphMX, kanMX, natMX, HIS3, URA3 medium: SC-ura
JC0044	YZy313 YZy363 (δ -integrated $P_{TDH3}-ILV2$, $P_{PGK1}-ILV3$, $P_{TEF1}-CoxIVMLS-Ll_adhA^{RE1}$, $P_{TDH3}-CoxIVMLS-ARO10$, $P_{TEF1}-ILV5$), CEN URA3 plasmid ($P_{TDH3}-ILV6-T_{ADH1}$) markers: hphMX, HIS3, ShBle, URA3 medium: SC-ura
JC0045	YZy314 YZy363, CEN URA3 plasmid ($P_{TDH3}-ILV6^{V110E}-T_{ADH1}$) markers: hphMX, HIS3, ShBle, URA3 medium: SC-ura
JC0050	YZy454 YZy452, CEN URA3 plasmid ($P_{TDH3}-Ll_ilvD-T_{ADH1}$) markers: hphMX, kanMX, HIS3, ShBle, URA3 medium: SC-ura
JC0051	YZy469 YZy452, CEN URA3 plasmid ($P_{TDH3}-Ll_ilvD^{I433V}-T_{ADH1}$) markers: hphMX, kanMX, HIS3, ShBle, URA3 medium: SC-ura

What the three axes buy together. The six span three pathway states, native flux in Axis 1, mitochondrial in Axis 2 and cytosolic in Axis 3, three genetic backgrounds, and roughly a threefold measured range in titer, on one carbon source and one assay format. Axis 1 carries no engineered route at all: both of its strains make isobutanol through endogenous valine biosynthesis and the Ehrlich pathway, with $bat1\Delta$ $bat2\Delta$ keeping α -ketoisovalerate from being transaminated away. The plasmid that separates its two strains carries *ILV6*, a regulatory subunit rather than a pathway enzyme, so the pair varies feedback control and not pathway capacity. Each pair isolates one variable, so a transcriptional difference within a pair has one candidate cause. Adding JC0044 is what makes Axis 2 a pair rather than a single high point, and without it JC0045 has no same-background comparator anywhere in the set.

4 Held back, and why $\times$

Table 4: The six held back from the first run, with the reason in each case. Four are held back because they make isopentanol rather than isobutanol. The other two make isobutanol but cannot share a run with the selected six: one produces only on galactose, and one needs a light schedule. The key below the table says how each titer is known.

id	strain	product	titer (mg/L)	reason
JC0049	YZy452	isobutanol	310 $\pm$ 15	galactose only: its chromosomal <i>Ll_ilvD</i> sits behind P_{GAL10} , so carbon source would confound every comparison in a shared run
JC0053	YZy502	isobutanol	not reported	triple <i>pdv</i> Δ with <i>PDC1</i> restored only under a blue-light promoter, so growth and production are separate light phases and a light-shift response would sit on top of the production response
JC0046	YZy148	isopentanol	not reported	makes isopentanol, not isobutanol
JC0047	YZy148 + LEU4 WT	isopentanol	$\sim 201^\dagger$	makes isopentanol, not isobutanol
JC0048	YZy148 + LEU4dS547	isopentanol	842 $\pm$ 23	makes isopentanol, not isobutanol
JC0054	SHy134	isopentanol	not reported	makes isopentanol, not isobutanol

bold stated in the paper as a number; † derived by dividing a stated fold change, so approximate and without uncertainty

JC0049 produces only on galactose. YZy452 carries its chromosomal *Ll_ilvD* behind P_{GAL10} , so the enzyme is repressed on glucose and the strain makes essentially no isobutanol there. Its 310 ± 15 mg/L is the highest stated isobutanol titer on hand, and it is a galactose number, so putting it in a glucose run would make carbon source a second

difference in every comparison involving it. Its genetics are already represented: it is the plasmid-free parent of JC0050 and JC0051, which supply the same background on glucose.

JC0053 needs a light schedule. YZy502 is *pdcl* Δ *pdcs* Δ *pdcs* Δ with *PDC1* restored under P_{C120}, which is induced by blue light, and the first enzyme of its cytosolic pathway under a circuit that is repressed by blue light and active in darkness. Growth and production therefore occupy different light conditions. A single-cell measurement taken across that transition would carry a light-shift response layered on the production response, and the paper reports no titer for YZy502 itself, only for its plasmid-bearing derivative. The strain is worth its own experiment with light as the variable, which is a different design from this one.

Four strains make the wrong product. JC0046, JC0047, JC0048 and JC0054 make isopentanol. JC0046 through JC0048 are otherwise the cleanest series in the whole set, being one parent carrying three plasmid states across a roughly fourfold titer range, and JC0048 holds the highest titer on hand at 842 ± 23 mg/L. They become the obvious first choice if isopentanol ever takes priority.

5 Before the strains are grown ✗

Three items, in the order they would block work.

JC0052 needs an empty vector, and the source paper shows why. Five of the six selected strains carry a *URA3* CEN plasmid and are held on SC-ura. JC0052 carries none, and the CEN.PK2-1C parent is *ura3-52*, so JC0052 needs uracil in its medium. Growing all six in medium containing uracil would remove selection from the other five, and a CEN plasmid is lost without selection, so by the end of a 48 h fermentation the population would carry plasmid-free cells. A single-cell measurement resolves that as a distinct subpopulation, which is a confound in a run designed to compare plasmid states.

The source paper met the same problem and solved it by transforming the plasmid-free strain with an empty vector: Fig. 3b reports YZy91 both with and without one, and YZy453 is YZy452 carrying empty pYZ125. Doing the same here means one transformation of JC0052 with pYZ125, a CEN *URA3* plasmid carrying P_{TDH3} and a multiple cloning site and no cargo, after which all six run on SC-ura. That strain is not among the twelve, so the transformation has to happen before the run. Skipping it leaves Axis 1 carrying a medium difference on top of its genetic one, and the two cannot be separated afterwards.

Axis 1 compares a deletion to an overexpression, not to a wild type. JC0043 supplies *ILV6* from P_{TDH3}, so the pair JC0052 against JC0043 is *ilv6* Δ against *ILV6* overexpressed. Neither strain has *ILV6* at its native level. A comparison against knockout-collection data, where the reference is a wild type carrying one chromosomal copy under its own promoter, has no matching strain in this set: the strains with *ILV6* intact, JC0044 and JC0045, sit in a different background and also carry a plasmid copy. Whether that gap matters is worth settling before the run, because closing it means building an isogenic *ILV6* wild type and that is a construction rather than a selection.

Transgene copy number is per-isolate on four of the six. JC0044, JC0045, JC0050 and JC0051 carry δ -integrated cassettes, so their copy number was fixed when the isolate was picked and is not specified by the construct. The source paper measured one to two copies by digital droplet PCR in the strains it checked, which were the sorted mitochondrial-pathway isolates rather than these four. If expression level of a pathway gene enters the analysis, these four want their own ddPCR first, since a twofold difference in copy number is indistinguishable from a twofold difference in regulation once the reads are counted.